STEP-BY-STEP BUILD GUIDE · NOVEMBER 2026

Building the Techno-Consultant Projects (Category A)

Flagship A: AI Transformation Blueprint for a mid-size Indian bank, with A1 Diagnostic, A2 Cost & ROI Calculator and A3 AI Value Dashboard

Tested Python, verified Excel formulas, DAX measures, templates and worked examples
12-week plan · traceability to every promised deliverable · interview answers

Prepared for

MBA (Business Analytics) candidate targeting Big 4 and top consulting GenAI roles in India

Kestrel Bank is a fictional client. CFPB complaint data is a public US stand-in; A3 data is simulated.
Facts are cited to a numbered Sources list and dated 7 November 2026. Prices change often.
Regulatory summaries support project design and are not legal advice.

Contents

How to Use This Guide . . . 3

The one-line objective of each project . . . 3

How each step is laid out . . . 3

0 Before You Start: Set-up, Client and Data . . . 4

A1 Complaint Process Diagnostic (Lean Six Sigma DMAIC) . . . 7

Define (week 1) . . . 7

Measure (week 2) . . . 8

Analyse (week 2) . . . 10

Improve and Control (week 3) . . . 12

A2 GenAI Cost and ROI Calculator (Excel) . . . 14

A3 AI Value Dashboard (Power BI) . . . 19

A Flagship A — AI Transformation Blueprint . . . 25

Final Check: Does It Meet the Objectives? . . . 34

Deliverables traceability . . . 34

Interview questions and where your answer is . . . 34

Quality checklist before you show it . . . 35

Common mistakes that cost credibility . . . 35

Sources . . . 36

START HERE

How to Use This Guide

This guide walks you through building the four **Category A (techno-consultant)** projects, one step at a time: Flagship A and its three supporting projects A1, A2 and A3. Every step says **what to do**, **how to do it** and **what you should have at the end**. Code and formulas in this guide were run and checked on test data before they were written down, and the business-case numbers were rebuilt in a spreadsheet engine to confirm the formulas.

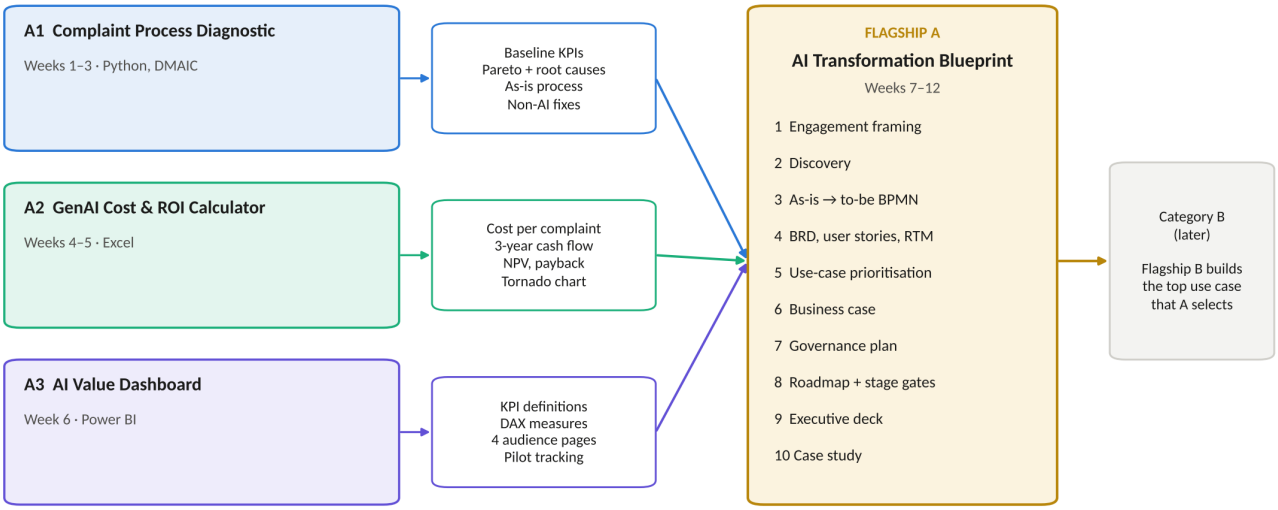


Figure 1. How the four projects connect. A1, A2 and A3 each produce an input that Flagship A needs. Flagship A then picks the use case that Flagship B (Category B, not covered here) will build.

The one-line objective of each project

Project	The question it answers	Main tool	Weeks
A1 Complaint Process Diagnostic	Where are time and quality lost in complaint handling, and which fixes need no AI at all?	Python (Pandas, Seaborn), DMAIC	1–3
A2 GenAI Cost and ROI Calculator	What does the AI cost per complaint, what does it return over three years, and which assumption matters most?	Excel	4–5
A3 AI Value Dashboard	Once the AI is live, how will each leader know whether it is working?	Power BI	6
Flagship A AI Transformation Blueprint	Where should the bank use AI in complaint handling, what will it return, and how do we roll it out safely?	PowerPoint, Visio / draw.io, Excel, MS Project	7–12

How each step is laid out

- **Step heading** in the project's colour: **A1 blue**, **A2 green**, **A3 purple**, **Flagship gold**.
- **Code boxes** are copy-ready. Python was tested with pandas 3.0, scikit-learn 1.9 and seaborn 0.13.
- **Blue boxes** are tips, **orange boxes** are warnings (things that will cost you credibility if you get them wrong), **green boxes** are the "done when" checklist at the end of each project.
- Numbers in brackets like [1] point to the Sources list at the end.

> **The consultant's rule for this whole guide**
>
> Every number you show must be one of three things: **measured** (from data), **sourced** (from a named document) or **assumed** (written in your assumptions register with a reason). Label which one it is. Interviewers at Big 4 firms probe numbers first; a labelled assumption is a strength, an unlabelled one is a red flag.

WEEK 0 · ABOUT ONE DAY

0 Before You Start: Set-up, Client and Data

STEP 0.1 Install the tools

Tool	Used in	Cost	Notes
Python 3 + Jupyter (Anaconda or VS Code)	A1, A3 data generator	Free	Install pandas, matplotlib, seaborn, scikit-learn: `pip install pandas matplotlib seaborn scikit-learn`
Microsoft Excel (desktop)	A2, Flagship A	Usually via college Microsoft 365	Needs Data Table and Scenario features (desktop Excel, not the web app).
Power BI Desktop	A3	Free	Windows only. On a Mac, use a Windows virtual machine or a college lab PC.
draw.io (diagrams.net) or Visio	A1, Flagship A	draw.io free	draw.io has BPMN shapes; save the .drawio file and export PNG.
MS Project, ProjectLibre or Excel	Flagship A roadmap	ProjectLibre free	Reuse your Project NOVA (#5) skills here.
PowerPoint or Google Slides	Flagship A deck	Free / college licence	16:9, one message per slide.

STEP 0.2 Create the folder structure

Use the same structure on your laptop and in a GitHub repository or a shared Google Drive folder. Interviewers notice tidy, numbered folders: it is what client deliverables look like.

Folder structure

```
complaints-ai-blueprint/
├── 00_admin/            assumptions_register.xlsx, sources.md, timeline.mpp
├── 01_A1_diagnostic/
│   ├── data/            raw/ (CFPB download, never edited)   clean/
│   ├── notebooks/       a1_step1_load.py ... a1_step4_narratives.py
│   └── outputs/         baseline_kpis.csv, pareto_issues.png, fishbone.png, as_is.drawio
├── 02_A2_business_case/ A2_GenAI_Cost_ROI.xlsx, A2_one_pager.pdf
├── 03_A3_dashboard/     ai_event_log.csv, baseline_targets.csv, A3_AI_Value.pbix, A3_pages.pdf
├── 04_FlagshipA/        discovery/, process/, requirements/, governance/, roadmap/, deck/
└── README.md            what this is, how to navigate, 5 key findings
```

STEP 0.3 Write the client brief (your fictional client)

All four projects use one fictional client so the story stays consistent. Write this half-page brief first and keep it in `00_admin`. Always say clearly that the client is fictional and that the public data is a stand-in.

Item	What to write (example)
Client	**Kestrel Bank** (fictional), a mid-size Indian private-sector bank with retail, cards and digital banking.
Situation	Complaint volumes are rising with digital adoption. Leadership wants to know whether GenAI can cut handling time and reduce escalations to the RBI Ombudsman without adding regulatory risk.
Ask to you	“Tell us where AI should and should not be used in complaint handling, what it is worth, and how to roll it out safely.”
Scale (assumed)	About 40,000 complaints a month handled by about 120 agents (allowing for breaks, training and leave) across four teams (Cards, Savings & Current, Digital Banking, Branch Escalations).
Constraints	Data must stay in India where possible; DPDP Act and RBI rules apply; no change to core banking in year one; budget approval needs payback inside 24 months.
Data stand-in	Public CFPB complaint data (US) is used for the *shape* of the problem — issue mix, response patterns, narrative themes. Indian-bank volumes, costs and times are labelled assumptions [1].

Where to get realistic Indian numbers

Indian banks publish a summary of customer complaints received and complaints from the Ombudsman in the notes to accounts of their **annual reports**. Look at two or three mid-size banks to sense-check the 40,000-per-month assumption, then cite what you used. Talking to two or three people who work in bank operations (alumni, LinkedIn) is even better — see Flagship A, Step 2.

STEP 0.4 Start the assumptions register

A simple Excel table you will add to in every project. It is the single most useful file in the whole package because it lets anyone challenge your numbers line by line.

ID	Assumption	Value	Type	Reason / source	Used in	Confidence
AS-01	Complaints per month	40,000	Assumed	Mid-size bank scale; check annual reports	A2, A3	Medium
AS-02	Average handle time today	25 min	Assumed	To validate in interviews; CFPB has no handling time	A2	Low
AS-03	Loaded agent cost	₹400 / hour	Assumed	Salary + overheads + seat cost	A2, A3	Medium
AS-04	Escalation rate to Ombudsman	3%	Assumed	To validate; annual-report disclosures help	A2	Low
AS-05	Claude Sonnet 5.5 price	$2 in / $10 out per 1M tokens	Sourced	Anthropic pricing page, Nov 2026 [11]	A2	High

STEP 0.5 Download the complaint data

1. Go to the CFPB Consumer Complaint Database page [1] and download the full CSV (it is large — over a gigabyte once unzipped). Save it unchanged in `01_A1_diagnostic/data/raw/`.
2. Read the field reference [4] and keep it open. The fields you will use are: Date received, Product, Sub-product, Issue, Sub-issue, Consumer complaint narrative, Company response to consumer, Timely response?, Submitted via, Date sent to company, Complaint ID.
3. For complaint text: since 14 August 2026 the CFPB no longer publishes narratives in the live database [2]. Narratives received from December 2011 to 14 August 2026 are in the FOIA Reading Room **Narratives Archive** [3]. Download the archive file(s) covering your time window, join them to the main file on Complaint ID, and use the narrative column from there.
4. Write a one-paragraph **data note** in `sources.md`: what you downloaded, the date, and the limitations below.

Three limitations to state up front (this is what a careful consultant does)

1. No resolution date. The data has the date the complaint was received and the date it was sent to the company, plus a "Timely response?" flag. It does not record when the complaint was resolved. So resolution time and cost per complaint cannot be measured from this data; they are assumptions you validate in interviews.

2. US data, Indian client. Use it for the mix of issues, response outcomes and themes in complaint text, not for volumes or costs.

3. Unverified, consumer-written text. Narratives are the customer's side only, and private details are masked as XXXX.

STEP 0.6 Plan the 12 weeks

Week	Project	Main output by end of week
1	A1 Define	Problem statement, SIPOC, CTQ tree, data loaded and filtered
2	A1 Measure + Analyse	Baseline KPI table, Pareto chart, 100 narratives tagged, fishbone
3	A1 Improve + Control	As-is process map, non-AI fixes, AI-opportunity tags, control plan, A1 one-pager
4	A2 Build	Inputs, unit cost and cash-flow sheets working; hand-check passes
5	A2 Analyse	Scenarios, tornado chart, one-page summary
6	A3	Data model, DAX measures, four pages, PDF export

Week	Project	Main output by end of week
7	Flagship A	Steps 1–2: framing, stakeholder map, interviews, pain-point register
8	Flagship A	Steps 3–4: to-be process, BRD, user stories, RTM
9	Flagship A	Steps 5–6: use-case scoring, matrix, business case
10	Flagship A	Step 7: governance plan, risk register, RACI
11	Flagship A	Steps 8–9: roadmap with stage gates, executive deck
12	Flagship A	Step 10: case study, README, LinkedIn post, mock interview

SUPPORTING PROJECT · WEEKS 1–3

A1 Complaint Process Diagnostic (Lean Six Sigma DMAIC)

A1 at a glance	
Objective	Find where complaint resolution time and quality are lost, and which fixes need no AI.
Builds on	Portfolio #6 Bank Complaint DMAIC — same method, now with real public data and a sharper AI lens.
Deliverables	Problem statement and charter · SIPOC · CTQ tree · baseline KPI table · Pareto chart · tagged narrative sample · fishbone and 5 Whys · as-is process map with step tags · list of non-AI fixes · AI-opportunity list · control plan · one-page summary.
Feeds	Flagship A Steps 2, 3 and 5 (pain points, as-is process, use-case long list); A2 inputs; A3 baselines.
Skills shown	Structured problem solving, Pandas EDA, root-cause analysis, process mapping, "is this even an AI problem?" judgement.

DMAIC has five phases: **Define** the problem, **Measure** today's performance, **Analyse** the root causes, **Improve** the process, **Control** the gains. Follow them in order — the order is the point.

Define (week 1)

STEP 1 Write the problem statement and project charter

A good problem statement describes the pain, where it happens and how big it is — with no solution in it. Fill the brackets once you have the Measure numbers, then come back and finish it.

> **Example problem statement**
>
> "Kestrel Bank handles about 40,000 complaints a month *[assumption AS-01]*. An estimated [12]% are not resolved within the 30 days after which customers may go to the RBI Ombudsman *[assumption]*, and about 3% are escalated *[AS-04]*. In the public stand-in data, [7] of [N] issue types account for 80% of complaints and [31]% of complaints end with relief for the customer, suggesting many complaints have merit and are avoidable. The goal of this diagnostic is to find the root causes and the fixes that need no new technology."

Under the Reserve Bank – Integrated Ombudsman Scheme, 2026 (in force from 1 July 2026) [5], a customer can approach the RBI Ombudsman if the bank has not replied within 30 days, or has rejected or unsatisfactorily resolved the complaint [6]. That makes "resolved within 30 days" the natural headline metric for an Indian bank.

Charter field	What to write
Problem	The statement above.
Goal (SMART)	e.g. "Identify fixes that could raise resolution within 30 days from [88]% to [95]% within 12 months."
In scope	Retail complaints: cards, savings and current accounts, digital banking; all intake channels.
Out of scope	Fraud investigations, loan collections, legal disputes, core-banking changes.
Stakeholders	Head of Customer Service (sponsor), Principal Nodal Officer, complaint team leads, IT, Compliance.
Timeline	3 weeks: Define (wk 1), Measure + Analyse (wk 2), Improve + Control (wk 3).

STEP 2 Draw the SIPOC

SIPOC is a one-table, high-level view of the process: who supplies inputs, the 5–7 main steps, the outputs and who receives them. It stops you from getting lost in detail too early.

Suppliers	Inputs	Process (high level)	Outputs	Customers
Customers; branches; call centre; RBI CMS portal	Complaint (text, channel, product); account and transaction data; policy documents	1 Receive and log 2 Acknowledge 3 Classify and route 4 Investigate 5 Decide outcome 6 Reply and close 7 Learn (root-cause review)	Reply letter / email; refund or fix; closed case; MIS reports	Customer; Principal Nodal Officer; Compliance; RBI (for Ombudsman cases)

STEP 3 Turn the voice of the customer into CTQs

CTQ means *critical to quality*: a measurable feature the customer cares about. Start from what customers say (read 20 narratives now), then turn each need into a metric and a target.

Customer need (in their words)	Driver	CTQ metric	Target (assumed)	Measurable in CFPB data?
“Fix it quickly”	Speed	% resolved within 30 days	≥ 95%	No — use “Timely response?” as a proxy
“Tell me you got it”	Communication	% acknowledged within 1 working day	≥ 98%	Partly — received → sent-to-company lag
“Don’t make me repeat myself”	Effort	First-contact resolution %	≥ 70%	No — interview / assumption
“Give me a fair outcome”	Fairness	Reopen rate; relief rate by issue	Reopen < 5%	Relief rate: yes

Measure (week 2)

STEP 4 Load and filter the data

The full file is too big to open in Excel, so read it in chunks and keep only bank products in a 24-month window. Product names changed over time (for example “Credit card” and “Credit card or prepaid card”), so match on text with `str.contains` instead of exact names.

a1_step1_load.py

```
import pandas as pd

RAW = "complaints.csv"              # the full CFPB download (large) — read it in chunks
PRODUCTS = r"Checking or savings account|Credit card"   # bank products only
START, END = "2024-09-01", "2026-08-31"                  # a 24-month window

keep = []
for chunk in pd.read_csv(RAW, chunksize=200_000, dtype=str):
    chunk["Date received"] = pd.to_datetime(chunk["Date received"],
                                            format="mixed", errors="coerce")
    mask = (chunk["Product"].str.contains(PRODUCTS, case=False, na=False)
            & chunk["Date received"].between(START, END))
    keep.append(chunk[mask])

df = pd.concat(keep, ignore_index=True)
df.to_csv("bank_complaints_24m.csv", index=False)
print(df.shape)
print(df["Product"].value_counts())
```

Check before moving on

Print `df["Product"].value_counts()` and make sure only the products you want are there. If you joined narratives from the archive, check how many rows have text — on real data a minority of complaints include a narrative, because consumers had to consent to publication.

STEP 5 Clean the data and calculate the baseline KPIs

Two rules: **remove exact duplicates on Complaint ID**, and **flag, never silently fix** anything odd (for example a "sent to company" date before the "received" date). Flags show the reviewer you checked.

a1_step2_clean_kpis.py

```
import pandas as pd

df = pd.read_csv("bank_complaints_24m.csv", dtype=str)

# 1. Types
for col in ["Date received", "Date sent to company"]:
    df[col] = pd.to_datetime(df[col], format="mixed", errors="coerce")

# 2. Duplicates: one row per Complaint ID
before = len(df)
df = df.drop_duplicates(subset="Complaint ID", keep="last")
print("Duplicates removed:", before - len(df))

# 3. Derived fields (flags, never silent fixes)
df["month"] = df["Date received"].dt.to_period("M").astype(str)
df["routing_lag_days"] = (df["Date sent to company"] - df["Date received"]).dt.days
df["flag_bad_dates"] = df["routing_lag_days"].isna() | (df["routing_lag_days"] < 0)
df["timely"] = df["Timely response?"].eq("Yes")
df["relief"] = df["Company response to consumer"].isin(
    ["Closed with monetary relief", "Closed with non-monetary relief"])
df["has_narrative"] = df["Consumer complaint narrative"].fillna("").str.strip().ne("")
```

a1_step2_clean_kpis.py (continued)

```
# 4. Baseline KPI table
months = df["month"].nunique()
kpis = {
    "Complaints in window": len(df),
    "Average complaints per month": round(len(df) / months),
    "Timely response rate (%)": round(100 * df["timely"].mean(), 1),
    "Median routing lag (days)": df.loc[~df["flag_bad_dates"], "routing_lag_days"].median(),
    "Share closed with relief (%)": round(100 * df["relief"].mean(), 1),
    "Share submitted via web (%)": round(100 * df["Submitted via"].eq("Web").mean(), 1),
    "Share with a narrative (%)": round(100 * df["has_narrative"].mean(), 1),
    "Rows with bad dates (flagged)": int(df["flag_bad_dates"].sum()),
}
baseline = pd.Series(kpis, name="Baseline")
print(baseline)
baseline.to_csv("baseline_kpis.csv")
df.to_csv("complaints_clean.csv", index=False)

# 5. Outcome mix and relief rate by issue (where the bank was most often at fault)
print(df["Company response to consumer"].value_counts(normalize=True).round(3))
relief_by_issue = (df.groupby("Issue")
                     .agg(complaints=("Complaint ID", "count"), relief_rate=("relief", "mean"))
                     .sort_values("complaints", ascending=False))
relief_by_issue["relief_rate"] = (100 * relief_by_issue["relief_rate"]).round(1)
print(relief_by_issue.head(10))
relief_by_issue.to_csv("relief_by_issue.csv")
```

What each KPI tells you, and how you will use it later:

KPI	What it means	Used in
Average complaints per month	Volume and trend. Look for seasonality and growth.	Problem statement; shape of the A2 volume assumption
Timely response rate	Share where the company responded within the CFPB's 15-day window. Closest proxy for speed.	CTQ baseline; A3 baseline
Median routing lag	Days from received to sent-to-company. Usually 0–2 days; shows intake is not the bottleneck.	As-is map timings
Share closed with relief	Monetary or non-monetary relief: a proxy for complaints where the customer had a point.	Root cause; avoidable-complaint estimate
Relief rate by issue	Which issues most often end with the bank putting something right.	Pareto follow-up; non-AI fixes

KPI	What it means	Used in
Channel mix; share with narrative	Where complaints come from; how much text you have to analyse.	SIPOC; Step 7

> **Do not do this**
>
> Do not report an “average resolution time” from CFPB data — the data has no resolution date. Say: “Resolution time is not in the public data; the 25-minute handle time and 30-day resolution rate are assumptions to be validated in interviews.” That sentence builds trust.

Analyse (week 2)

STEP 6 Build the Pareto chart

a1_step3_pareto.py

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

df = pd.read_csv("complaints_clean.csv", dtype=str)
counts = df["Issue"].value_counts()
top = counts.head(12)
cum_pct = 100 * top.cumsum() / counts.sum()   # cumulative share of ALL complaints

fig, ax = plt.subplots(figsize=(11, 5))
sns.barplot(x=top.values, y=top.index, ax=ax, color="#2a78d6")
ax.set_xlabel("Number of complaints")
ax.set_ylabel("")
ax.set_title("Top complaint issues (Pareto)")
for i, (v, c) in enumerate(zip(top.values, cum_pct.values)):
    ax.text(v, i, f"  {c:.0f}% cumulative", va="center", fontsize=8)
plt.tight_layout()
plt.savefig("pareto_issues.png", dpi=200)

# How many issues make up 80% of complaints?
all_cum = 100 * counts.cumsum() / counts.sum()
n80 = int((all_cum < 80).sum()) + 1
print("Issues needed to reach 80% of complaints:", n80, "of", len(counts))
```

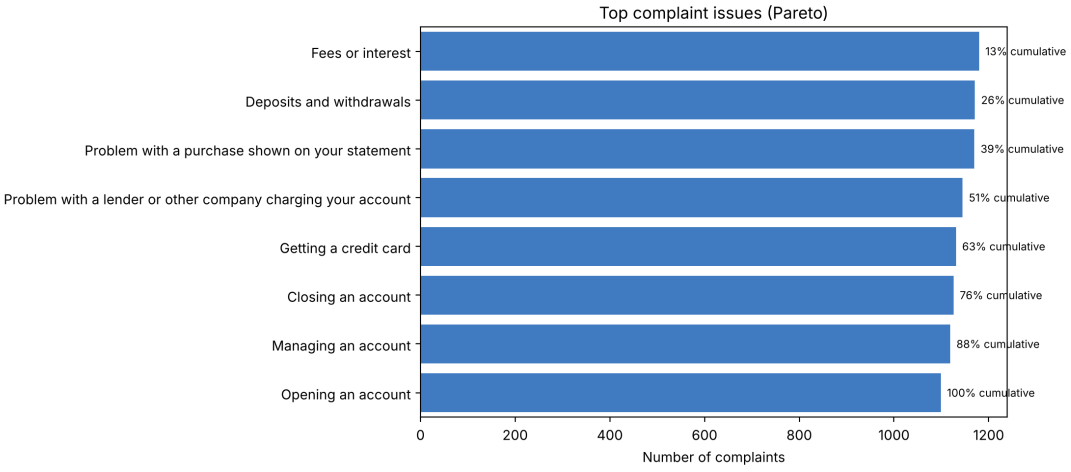


Figure 2. What the output looks like (shown here on synthetic test data, so the bars are flat). On the real data, write your 80/20 sentence from the printed result.

Write one sentence from the chart: *“[7] of [N] issue types account for 80% of bank complaints; the top three are [A], [B], [C].”* If the curve is flat, say so — it means there is no single fix, which is itself a finding. Then look at `relief_by_issue.csv`: an issue that is both **frequent** and has a **high relief rate** is your best candidate for root-cause work.

STEP 7 Read and tag 100 complaint narratives (the most valuable step)

Numbers tell you *where*; text tells you *why*. The script below saves a random sample of 100 narratives to a CSV. Open it in Excel and fill two columns by hand. Allow three to four hours. This is what separates a consultant from a dashboard builder.

Column	Allowed values	Tagging rule
root_cause	Process · Policy · People · System · Information · Customer understanding	Pick the one cause that, if fixed, would have prevented the complaint. Example: fee charged against stated policy → Policy; "was transferred five times" → Process.
ai_could_help	Triage · Knowledge · Draft · Summary · None	Ask: would an AI tool have made this faster or better *for the agent*? If the cause is a broken policy, the answer is None — fix the policy.

a1_step4_narratives.py

```
import pandas as pd
from sklearn.feature_extraction.text import TfidfVectorizer
from sklearn.cluster import KMeans

# Narratives come from the CFPB archive files, filtered the same way as Step 1
df = pd.read_csv("complaints_clean.csv", dtype=str)
text = df["Consumer complaint narrative"].fillna("").str.strip()
narr = df[text.ne("")].copy()

# A. 100 complaints to read and tag by hand (the most valuable part)
sample = narr.sample(n=min(100, len(narr)), random_state=42)
sample[["Complaint ID", "Product", "Issue", "Consumer complaint narrative"]].assign(
    root_cause="", ai_could_help="").to_csv("narrative_sample_to_tag.csv", index=False)

# B. Optional: find themes automatically (reuses your K-Means skills from project #2)
from sklearn.feature_extraction.text import ENGLISH_STOP_WORDS
stops = list(ENGLISH_STOP_WORDS | {"xxxx", "xx"})   # CFPB masks private details as XXXX
vec = TfidfVectorizer(stop_words=stops, max_features=5000, min_df=5,
                      token_pattern=r"(?u)\b[a-zA-Z]{3,}\b")   # words of 3+ letters only
X = vec.fit_transform(narr["Consumer complaint narrative"])
k = 6
km = KMeans(n_clusters=k, n_init=10, random_state=42).fit(X)
narr["theme"] = km.labels_
terms = vec.get_feature_names_out()
for i, centre in enumerate(km.cluster_centers_):
    top_words = [terms[j] for j in centre.argsort()[-8:][::-1]]
    print(f"Theme {i} ({(narr['theme'] == i).sum()} complaints):", ", ".join(top_words))
```

Part B is optional. It groups similar narratives into themes with TF-IDF and K-Means, reusing your Customer Segmentation (#2) skills. Read 5 complaints from each theme and give the theme a plain-English name. Use the themes to check that your manual tags did not miss anything big; do not present the clusters as the answer on their own.

STEP 8 Fishbone and 5 Whys

Put the effect (the problem) at the head of the fish and group causes under six headings. Use your tags from Step 7 and the Pareto, so every bone is backed by evidence. Draw it in draw.io or PowerPoint.

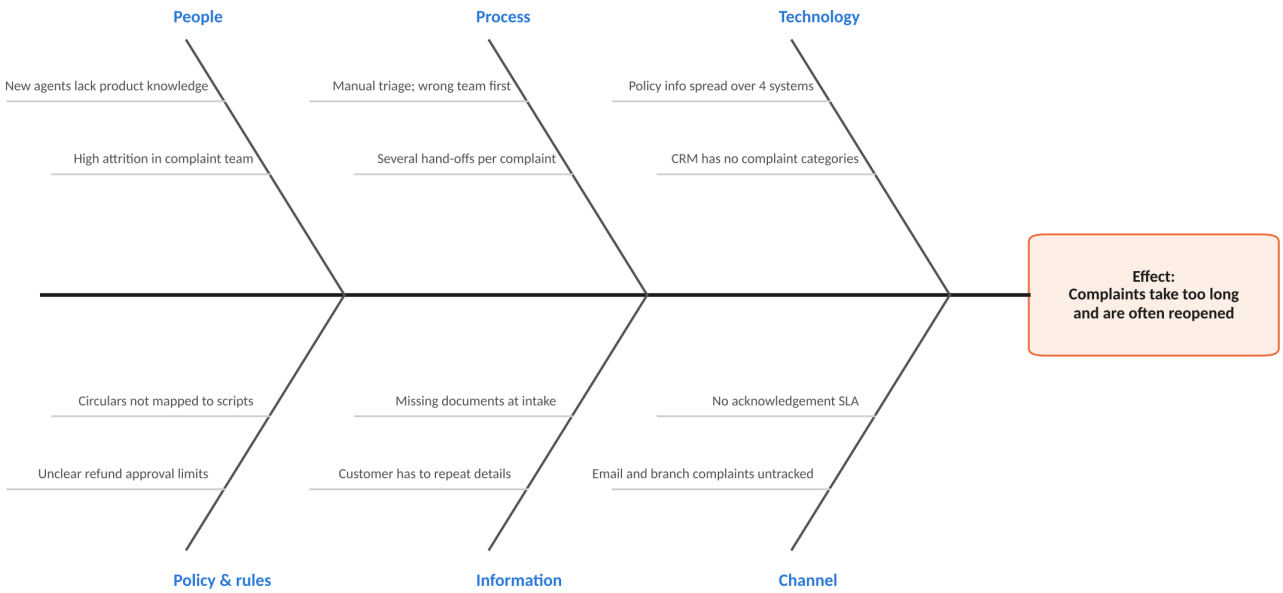


Figure 3. Example fishbone for Kestrel Bank. Replace these example causes with the ones your data and tags support.

Then pick the top two causes and ask "why?" until you reach something the bank can act on (usually 4–5 levels):

Why?	Answer (example)
Why are complaints slow to resolve?	Agents spend a long time finding the right policy and past cases.
Why does finding the policy take long?	Policies and RBI circulars sit in four systems with no single search.
Why are there four systems?	Each product team keeps its own documents; nobody owns the complaint knowledge base.
Why does nobody own it?	Complaint handling is measured on closure speed, not on knowledge quality.
Root cause	**No owner or process for complaint knowledge** → fix ownership first (non-AI); a knowledge Q&A tool then helps.

STEP 9 Map the as-is process

1. Draw swimlanes in draw.io: Customer, Channels + CRM, Complaint agent, Supervisor/QA, Other teams.
2. Add each step as a box. Mark hand-offs between lanes — every hand-off is a place where time is lost.
3. Write the time per step under each box and label it *measured* or *assumed*.
4. Tag each step: **H** human, **R** rule or workflow, **AI** AI call, **AG** agent. In the as-is map almost everything is H.
5. Circle the waste using the Lean list: waiting, rework, extra hand-offs, searching for information, re-entering data.

You will redraw this as the to-be process in Flagship A, Step 3, so save it as an editable file.

Improve and Control (week 3)

STEP 10 List the fixes that need no AI

This list is what makes interviewers trust your later AI recommendations. Aim for five to eight fixes.

Non-AI fix (example)	Root cause addressed	Effort	Owner
Acknowledge every complaint within 1 working day with an automatic message	No acknowledgement SLA	Low	Contact-centre ops
Standard complaint categories in the CRM (from your Pareto issue list)	CRM has no categories	Low	CRM team
Clear refund approval limits by grade, published to agents	Unclear approval limits	Low	Operations + Finance
Single owner for complaint knowledge; monthly policy-to-script review	Nobody owns knowledge	Medium	Principal Nodal Officer
Day-25 alert for open complaints (before the 30-day Ombudsman threshold)	Complaints drift past 30 days	Low	CRM team

Non-AI fix (example)	Root cause addressed	Effort	Owner
Fix the product issue behind the top high-relief complaint type	Repeat, avoidable complaints	Varies	Product owner

STEP 11 Tag where AI could help

Go back to the as-is map and your narrative tags. For each step, ask whether AI would help the *agent* do it faster or better, and what kind of AI. Keep this as a simple list; Flagship A Step 5 scores it properly.

As-is step	Pain	Possible AI help	Evidence
Classify and route	Wrong team first; re-routing	Classify complaint and suggest the team	Pareto + [x]% of tags = Triage
Investigate	Searching four systems for policy	Policy and RBI-rule Q&A with citations (RAG)	[x]% of tags = Knowledge
Reply	Writing long letters from scratch	Draft reply for the agent to edit	[x]% of tags = Draft
Escalation	Supervisor reads the whole history	Case summary	Interviews

STEP 12 Write the control plan

KPI	Definition	Target	Frequency	Owner	If off target...
Resolved within 30 days	Closed complaints with reply ≤ 30 days ÷ all closed	≥ 95%	Weekly	Ops head	Review day-25 alert list daily
Acknowledged within 1 day	Acknowledged ≤ 1 working day ÷ all received	≥ 98%	Daily	CRM team	Check auto-message rule
Reopen rate	Reopened within 30 days ÷ closed	< 5%	Monthly	QA lead	Sample 20 reopened cases
Escalation to Ombudsman	Ombudsman complaints ÷ all complaints	≤ 2.55%	Monthly	Nodal officer	Root-cause review of each case

STEP 13 Write the one-page A1 summary

- **Title as a finding**, not a topic: “Three issue types drive most complaints, and the biggest root cause is knowledge, not people.”
- Left half: Pareto chart and baseline KPI table. Right half: top three root causes, five non-AI fixes, four AI opportunities.
- Footer: data note (CFPB stand-in, no resolution date, assumptions labelled).

Done when...

- ☐ Problem statement and charter filled in with real numbers from your data
- ☐ SIPOC and CTQ table done
- ☐ `baseline_kpis.csv`, `relief_by_issue.csv` and the Pareto chart saved
- ☐ 100 narratives tagged; counts of root_cause and ai_could_help summarised
- ☐ Fishbone, 5 Whys, and as-is map with H / R / AI / AG tags
- ☐ Non-AI fix list, AI-opportunity list and control plan
- ☐ One-page summary exported to PDF; every number marked measured, sourced or assumed

SUPPORTING PROJECT · WEEKS 4–5

A2 GenAI Cost and ROI Calculator (Excel)

A2 at a glance	
Objective	Turn usage assumptions into cost per complaint, a three-year cash flow, NPV, payback and ROI — and show which assumption matters most.
Builds on	New project; uses your MBA finance and business-case skills.
Deliverables	Inputs sheet with Worst / Base / Best scenarios · unit-cost calculator · three-year cash flow · summary (NPV, payback, ROI) · tornado chart · one-page summary.
Feeds	Flagship A Step 6 (business case) and the A3 dashboard assumptions (FX, agent cost, realisation).
Skills shown	Token economics, prompt caching, model routing, TCO, NPV and payback, sensitivity analysis, honest benefit framing.

This workbook was rebuilt cell by cell and recalculated in a spreadsheet engine to confirm every formula below. If you copy the layout exactly, your numbers should match the check table in Step 6 to the rupee.

STEP 1 Plan the workbook and set the house rules

Sheet	Purpose
Inputs	Every assumption, in one place, with Worst / Base / Best columns, units and sources. The only sheet anyone should edit.
UnitCost	Cost per AI call and per complaint for each use case, and minutes saved per complaint.
CashFlow	Year 0 to Year 3: benefits, costs, net cash flow, cumulative cash flow, payback helpers.
Summary	NPV, payback, ROI and token share — the numbers that go on the slide.
Tornado	Low and high NPV for each driver, and the chart.

- **Inputs in blue font, formulas in black.** Anyone can see at a glance what they are allowed to change.
- **No typed numbers inside formulas.** Every number lives on the Inputs sheet and is referred to by name.
- **Units in every label**: “(INR/hr)”, “(min)”, “$/1M tokens”. Most spreadsheet errors are unit errors.

STEP 2 Build the Inputs sheet with a scenario switch

Type the scenario selector in cell B1 (1 = Worst, 2 = Base, 3 = Best). Put headers in row 3: *Input · Live value · Worst · Base · Best · Unit · Source*. Type the inputs from row 4. The **Live value** column picks the right scenario:

Inputs!B4 (Live value)

```
B4:  =CHOOSE(Scenario, C4, D4, E4)        copy down to B19
```

Naming a cell: click the cell, type the name in the Name Box (left of the formula bar), press Enter. Name B1 `Scenario` and each Live-value cell as in the table.

Row	Input	Name	Worst	Base	Best	Type
4	Complaints per month	Complaints_m	40,000	40,000	40,000	Assumed
5	Average handle time today (min)	AHT	25	25	25	Assumed
6	Loaded agent cost (INR/hr)	Cost_hr	400	400	400	Assumed
7	Target adoption	Adoption	55%	80%	90%	Assumed
8	Acceptance rate (AI output used, as-is or edited)	Acceptance	70%	85%	92%	Assumed
9	Realisation factor (share of saved time that becomes money)	Realisation	35%	50%	65%	Assumed
10	Escalation rate to Ombudsman	Esc_rate	3%	3%	3%	Assumed
11	Escalation reduction	Esc_cut	8%	15%	22%	Assumed

Row	Input	Name	Worst	Base	Best	Type
12	Cost per escalation (INR)	Esc_cost	5,000	5,000	5,000	Assumed
13	INR per USD	FX	88	88	88	Assumed — update on the day
14	Discount rate	Disc	12%	12%	12%	Assumed
15	Cache hit rate	Cache_hit	90%	90%	90%	Assumed
16	Cache write multiplier	Cache_write	1.25	1.25	1.25	Sourced [12]
17–19	Adoption ramp, Years 1 / 2 / 3	Ramp_Y1, _Y2, _Y3	40% / 85% / 100%	same	same	Assumed

Below the inputs add the price table (rows 21–23) and the cost lines (rows 25–36). Name the price block A22:D23 `Prices`, the build costs B26:B30 `Build_rng` and the run costs B33:B36 `Run_rng`.

Model (row)	Input $/1M tokens	Cached input $/1M	Output $/1M	Source
Claude Sonnet 5.5 (22)	2.00	0.10	10.00	Anthropic pricing, Nov 2026 [11]
Claude Haiku 5.5 (23)	0.10	0.01	0.50	Same; cached reads are 10% of the input price [12]

Build cost, one-off (rows 26–30)	INR	Run cost per year, excl. tokens (rows 33–36)	INR
Discovery and proof of concept	18,00,000	Cloud hosting, vector store, logging	9,00,000
MVP build and hardening	40,00,000	Monitoring and evaluation tooling	4,00,000
CRM and core-system integration	15,00,000	Run team (1 engineer + 0.5 product owner)	22,00,000
Security, DPDP and model-risk review	7,00,000	Quarterly re-testing and model upgrades	5,00,000
Training and change management	6,00,000		
Total build	**86,00,000**	**Total run**	**40,00,000**

> **Why these costs are realistic to show**
>
> People, integration and governance are most of the cost of an AI programme; tokens are a small slice. Your model will show tokens at about 8% of three-year cost. Saying this unprompted in an interview signals real commercial understanding.

STEP 3 Build the UnitCost sheet

One row per use case (rows 2–4), a total in row 5. Columns A–H are typed inputs; I–N are formulas. Fill row 2, then copy down.

Use case	Model	Coverage	Calls per complaint	Input tokens	Cacheable tokens	Output tokens	Minutes saved
Triage and routing	Claude Haiku 5.5	100%	1.0	1,200	800	60	2.5
Draft reply	Claude Sonnet 5.5	70%	1.5	6,000	2,000	450	6.0
Policy knowledge Q&A	Claude Sonnet 5.5	60%	1.0	8,500	3,000	500	4.0

Coverage is the share of complaints where the use case is used. *Minutes saved* is per covered complaint when the agent accepts the output. Both are assumptions to validate with agents in Flagship A, Step 2.

UnitCost formulas (row 2, copy to rows 3–4)

```
I2:  =VLOOKUP(B2, Prices, 2, FALSE)          input price  $/1M
J2:  =VLOOKUP(B2, Prices, 3, FALSE)          cached price $/1M
K2:  =VLOOKUP(B2, Prices, 4, FALSE)          output price $/1M
L2:  =((E2-F2)*I2 + F2*Cache_hit*J2 + F2*(1-Cache_hit)*I2*Cache_write + G2*K2) / 1000000
                                             cost per call (USD)
M2:  =L2*D2*C2                               cost per complaint (USD)
N2:  =H2*C2                                  minutes saved per complaint
M5:  =SUM(M2:M4)    name it Cost_per_complaint
N5:  =SUM(N2:N4)    name it Min_per_complaint
```

The cost-per-call formula in words: uncached input tokens at the full input price, plus cacheable tokens that hit the cache at the cheap cached price, plus cacheable tokens that miss the cache and are written to it at 1.25× the input price, plus output tokens at the output price — all divided by one million because prices are per million tokens.

> **Hand-check one row before you trust the sheet**
>
> Policy knowledge Q&A on Sonnet 5.5: 5,500 uncached × $2 = 11,000 · 2,700 cache hits × $0.10 = 270 · 300 cache writes × $2 × 1.25 = 750 · 500 output × $10 = 5,000. Total 17,020 ÷ 1,000,000 = **$0.01702 per call**. If your L4 does not show 0.01702, find the error now.

STEP 4 Build the CashFlow sheet

Columns B to E are Year 0 to Year 3; F is the total. Type the formula in column B, copy it across to E, except the four rows marked. Year 0 has a ramp of 0, so every benefit and token cost there is automatically zero.

Row	Line	Formula in column B (copy to C:E)
2	Ramp	B2: `0` · C2: `=Ramp_Y1` · D2: `=Ramp_Y2` · E2: `=Ramp_Y3`
3	Complaints handled with AI	`=Complaints_m*12*Adoption*B2`
4	Hours saved	`=B3*Min_per_complaint*Acceptance/60`
5	Labour value, gross (INR)	`=B4*Cost_hr`
6	Hard benefit (INR)	`=B5*Realisation`
7	Soft benefit: escalations avoided (INR)	`=B3*Esc_rate*Esc_cut*Esc_cost`
8	Full benefit (INR)	`=B6+B7`
9	Build cost (INR)	B9: `=SUM(Build_rng)` · C9:E9: `0`
10	Run cost excl. tokens (INR)	B10: `0` · C10:E10: `=SUM(Run_rng)`
11	Token cost (INR)	`=B3*Cost_per_complaint*FX`
12	Total cost (INR)	`=B9+B10+B11`
13	Net cash flow, hard only	`=B6-B12`
14	Net cash flow, full	`=B8-B12`
15	Cumulative, hard only	B15: `=B13` · C15: `=B15+C13` (copy to E)
16	Cumulative, full	B16: `=B14` · C16: `=B16+C14` (copy to E)
F	Total column (rows 3–14)	`=SUM(B3:E3)`, copied down

> **Why “hard” and “full” benefits are kept apart**
>
> **Hard benefit** is the part of saved agent time that turns into money (fewer new hires, less overtime, less outsourcing) — hence the realisation factor. Saved time is not cash until someone decides what to do with it.
>
> **Soft benefit** here is escalations avoided. It is real but less certain, so it is shown separately. A CFO will often approve on hard benefits and treat the rest as upside — show both and let them choose.

STEP 5 Add NPV, payback and ROI on the Summary sheet

Payback helpers and Summary formulas

```
CashFlow!C17:  =IF(AND(B15<0, C15>=0), 0+(-B15)/C13, "")      payback helper, hard
CashFlow!D17:  =IF(AND(C15<0, D15>=0), 1+(-C15)/D13, "")
CashFlow!E17:  =IF(AND(D15<0, E15>=0), 2+(-D15)/E13, "")
CashFlow!C18:E18:  same three formulas using rows 16 and 14              payback helper, full

Summary!B1  NPV, hard only     =NPV(Disc, CashFlow!C13:E13) + CashFlow!B13
Summary!B2  NPV, full          =NPV(Disc, CashFlow!C14:E14) + CashFlow!B14
Summary!B3  Payback (yrs) hard =IF(COUNT(CashFlow!C17:E17)=0, "Not within 3 years", SUM(CashFlow!C17:E17))
Summary!B4  Payback (yrs) full =IF(COUNT(CashFlow!C18:E18)=0, "Not within 3 years", SUM(CashFlow!C18:E18))
Summary!B5  3-year ROI, full   =(CashFlow!F8 - CashFlow!F12) / CashFlow!F12
Summary!B6  Token share        =CashFlow!F11 / CashFlow!F12
Summary!B7  Cost per complaint (INR)  =Cost_per_complaint * FX
```

- **NPV:** Excel's NPV function assumes the first value is one year away, so Year 0 (the build cost) is added outside it. This is the most common NPV mistake in student models.
- **Payback:** each helper cell checks whether cumulative cash turns positive in that year and, if so, how far into the year (assuming cash arrives evenly). Multiply years by 12 for months.

STEP 6 Check your build against these results

With the Base scenario (B1 = 2), your workbook should show exactly these values. Then switch B1 to 1 and 3.

Result	Worst (B1 = 1)	Base (B1 = 2)	Best (B1 = 3)
Cost per complaint (all three use cases)	₹2.12 ($0.0241)	₹2.12 ($0.0241)	₹2.12 ($0.0241)
Hours saved in Year 3	—	49,504	—
NPV, full view	−₹68.0 lakh	**₹127.5 lakh**	₹324.6 lakh
NPV, hard benefits only	−₹123.3 lakh	−₹23.4 lakh	₹75.6 lakh
Payback, full view	Not within 3 years	**1.50 years (18 months)**	1.05 years (13 months)
Payback, hard only	Not within 3 years	Not within 3 years	1.78 years (21 months)
3-year ROI, full view	−27%	86%	197%
Tokens as share of 3-year cost	5.8%	8.2%	9.1%

STEP 7 Build the tornado chart

A tornado chart shows which assumption moves NPV most. Use the Base scenario. For each driver, change the input to its low value, write down NPV (full), change it to its high value, write down NPV, then put the base value back. Record the results on the Tornado sheet.

Driver	Where to change it	Low → NPV	High → NPV
Adoption	Inputs!D7	50% → ₹11.4 lakh	95% → ₹185.5 lakh
Cost per escalation	Inputs!D12	₹2,500 → ₹52.1 lakh	₹8,000 → ₹218.0 lakh
Realisation factor	Inputs!D9	30% → ₹58.3 lakh	70% → ₹196.7 lakh
Minutes saved	UnitCost!H2:H4 (all three × 0.6 / × 1.3)	₹58.3 lakh	₹179.4 lakh
Agent cost per hour	Inputs!D6	₹300 → ₹84.3 lakh	₹500 → ₹170.7 lakh
Run cost	Inputs!B33:B36 (all × 0.75 / × 1.5)	₹151.5 lakh	₹79.5 lakh
Acceptance rate	Inputs!D8	65% → ₹86.8 lakh	95% → ₹147.8 lakh
Token prices	Inputs!B22:D23 (all × 0.5 / × 3)	₹134.6 lakh	₹99.0 lakh

Add two helper columns: *low − base* and *high − base* (in ₹ lakh). Sort the rows by swing (high minus low), largest first. Insert a **stacked bar chart** of the two helper columns: the downside bars extend left of zero and the upside bars right. Set series overlap to 100%, put the vertical axis in reverse order so the largest driver is at the top, colour downside orange and upside green, and add data labels showing the actual NPVs. Title the axis "Change in NPV vs base (₹ lakh)".

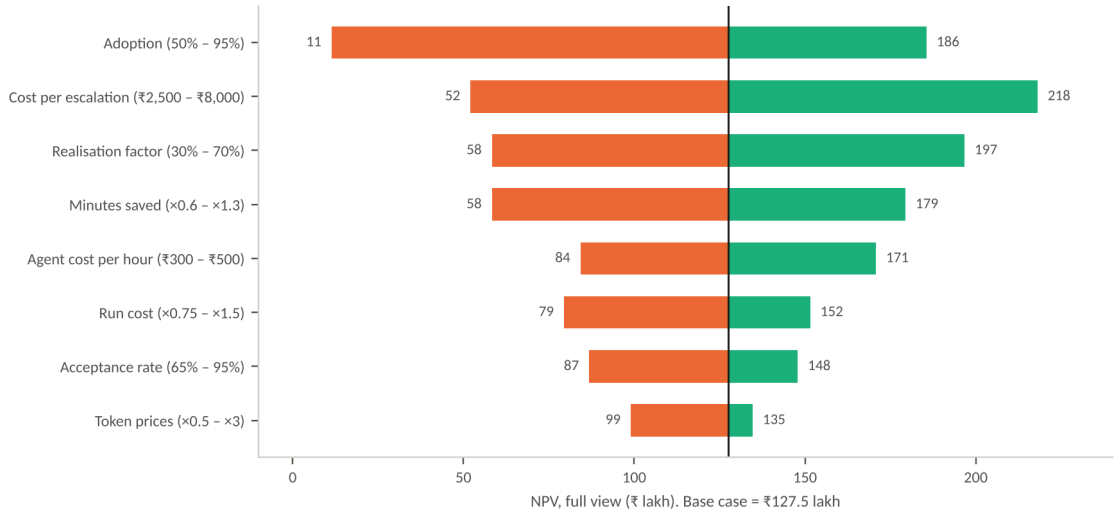


Figure 4. Tornado chart from the verified model. Adoption and the value of an avoided escalation matter most; token prices matter least — even tripling them leaves NPV at about ₹99 lakh.

Faster option: one-variable Data Tables

Data → What-If Analysis → Data Table can calculate NPV for a list of input values in one go. Two rules: the input cell must be on the **same sheet** as the Data Table, and it must be a typed value, not a formula. So point it at the Base column (column D) on the Inputs sheet, and build the tables on that sheet.

STEP 8 Read the results like a consultant and write the one-page summary

Your one-pager should lead with four messages. These are what the verified Base case says:

1. **AI is cheap per complaint; agent time is not.** About ₹2.12 of model usage per complaint, against about ₹167 of agent time (25 minutes at ₹400 an hour).
2. **The base case pays back in about 18 months — but only with the escalation benefit.** On labour savings alone it does not pay back within three years (NPV −₹23 lakh). The investment is a customer-and-regulatory case, not a cost-cutting case.
3. **Adoption is the biggest risk.** Below about 50% adoption the case barely breaks even, so the rollout plan must be built around agent adoption (training, workflow fit), not around the model.
4. **Tokens are about 8% of three-year cost.** People, integration and governance dominate. Switching to a cheaper model is not the main lever.

Message 2 sets up your Flagship A recommendation: **stage-gate the investment** and make Gate 2 depend on measured escalation reduction and adoption in the pilot. That is exactly what A3 will measure.

Interview-ready: two questions you will be asked

"Estimate the monthly LLM cost for 2,000 users." Users × queries per day × working days × cost per query. 2,000 × 10 × 22 × $0.017 ≈ $7,500 a month, about ₹6.6 lakh. Say your assumptions out loud as you go.

"How would you cut the cost?" (1) Route simple tasks to a small model (already done for triage); (2) cache the long shared prompt; (3) use the 50%-cheaper batch mode for work that can wait, such as nightly theme mining; (4) cap output length; and (5) the biggest lever — raise adoption, because fixed costs are spread over more complaints.

Done when…

- ☐ Inputs sheet with Worst / Base / Best, named ranges, units and a source for every row
- ☐ Hand-check of the knowledge Q&A row gives $0.01702 per call
- ☐ Base-case Summary matches the Step 6 table
- ☐ Tornado sheet and chart, sorted by swing
- ☐ One-page summary with the four messages and the hard-versus-full distinction
- ☐ Every assumption added to the assumptions register

SUPPORTING PROJECT · WEEK 6

A3 AI Value Dashboard (Power BI)

A3 at a glance	
Objective	Show leaders whether the AI is working — cost, quality, speed and adoption — each for the right audience.
Builds on	Portfolio #1 Sales Insights Dashboard (Power BI skills) and #3 Client Feedback Automation (data flowing from a tool into reports).
Deliverables	Star-schema data model · 8+ DAX measures with a KPI dictionary · four pages (CFO, COO, Risk, Sponsor) · "how to read this" notes · .pbix file and PDF export.
Feeds	Flagship A Steps 7–8 (monitoring in the governance plan; go / no-go evidence at each stage gate). Later, Flagship B's real logs.
Skills shown	Outcome-based KPI design, data modelling, DAX, audience-specific storytelling; PL-300 practice.

There is no real AI log yet, so you will **simulate the first 90 days of Phase 2** of the rollout (all four teams, 120 agents, three use cases). Say clearly on the cover page that the data is simulated. The point of A3 is the measurement design, which stays the same when real data arrives.

STEP 1 Generate the simulated AI event log

One row per AI run. Costs use the same prices as A2; minutes saved use the same per-use-case values. Run the script and you get two files: `ai_event_log.csv` (about 64,600 rows) and `baseline_targets.csv`.

a3_simulate_ai_log.py

```
import numpy as np
import pandas as pd

rng = np.random.default_rng(7)
N_DAYS, START = 90, "2026-01-05"          # a 90-day pilot (fictional)
PRICE = {                                  # USD per 1M tokens: input, cached input, output
    "Claude Sonnet 5.5": (2.00, 0.10, 10.00),
    "Claude Haiku 5.5":  (0.10, 0.01, 0.50),
}
# use case: (model, share of events, avg input tokens, avg output tokens, minutes saved)
USE_CASES = {
    "Triage & routing":     ("Claude Haiku 5.5",  0.45, 1_200,  60, 2.5),
    "Draft reply":          ("Claude Sonnet 5.5", 0.35, 6_000, 450, 6.0),
    "Policy knowledge Q&A": ("Claude Sonnet 5.5", 0.20, 8_500, 500, 4.0),
}
TEAMS = ["Cards", "Savings & Current", "Digital Banking", "Branch Escalations"]
AGENTS = 120                               # complaint-handling agents in the pilot
```

a3_simulate_ai_log.py (continued)

```
rows = []
event_id = 0
for day in pd.date_range(START, periods=N_DAYS, freq="D"):
    ramp = min(1.0, 0.25 + day.dayofyear / 60)     # adoption ramps up over the first weeks
    active = int(AGENTS * ramp)                          # agents using the copilot so far
    n = int(rng.poisson(900 * ramp) * (0.55 if day.dayofweek == 6 else 1))  # quiet Sundays
    for _ in range(n):
        uc = rng.choice(list(USE_CASES), p=[v[1] for v in USE_CASES.values()])
        model, _, inp, out, mins = USE_CASES[uc]
        tin = int(rng.normal(inp, inp * 0.15))
        tcache = int(tin * 0.35 * (rng.random() < 0.9))   # ~90% cache hit on shared prompt
        tout = max(20, int(rng.normal(out, out * 0.2)))
        p_in, p_cache, p_out = PRICE[model]
        cost = ((tin - tcache) * p_in + tcache * p_cache + tout * p_out) / 1e6
        outcome = rng.choice(["Accepted", "Edited", "Rejected"],
                             p=[0.55 + 0.15 * ramp, 0.30 - 0.10 * ramp, 0.15 - 0.05 * ramp])
        saved = {"Accepted": mins, "Edited": mins * 0.5, "Rejected": -0.5}[outcome]
        event_id += 1
        agent = int(rng.integers(1, active + 1))
        rows.append({
            "event_id": event_id,
            "timestamp": day + pd.Timedelta(minutes=int(rng.integers(8 * 60, 20 * 60))),
            "date": day.date(),
            "agent_id": f"AG{agent:03d}",
```

a3_simulate_ai_log.py (continued)

```
            "team": TEAMS[agent % 4],
            "use_case": uc,
            "model": model,
            "input_tokens": tin, "cached_tokens": tcache, "output_tokens": tout,
            "cost_usd": round(cost, 6),
            "latency_ms": int(rng.lognormal(np.log(1800 if "Sonnet" in model else 600), 0.35)),
            "agent_outcome": outcome,
            "minutes_saved": saved,
            "guardrail_flag": rng.choice(
                ["No flag", "PII blocked", "Low confidence", "Off-policy"],
                                         p=[0.965, 0.015, 0.015, 0.005]),
            "escalated_to_human": "Yes" if rng.random() < 0.06 else "No",
        })

log = pd.DataFrame(rows)
log.to_csv("ai_event_log.csv", index=False)

# Baseline (pre-AI) values from A1 + your assumptions; the dashboard compares against these
pd.DataFrame({
    "metric":   ["Avg handle time (min)", "Complaints per month",
                 "Loaded agent cost (INR/hr)",
                 "Resolved within 30 days (%)", "Escalated to Ombudsman (%)"],
    "baseline": [25, 40_000, 400, 88.0, 3.0],
    "target":   [19, 40_000, 400, 95.0, 2.55],
```

a3_simulate_ai_log.py (continued)

```
    "source":   ["Assumption (interviews)", "Assumption (A2 input)",
                 "Assumption (A2 input)",
                 "Assumption (A1 timely rate as proxy)", "Assumption (A2 input)"],
}).to_csv("baseline_targets.csv", index=False)

print(log.shape)
print(log.groupby("use_case")["cost_usd"].agg(["count", "mean", "sum"]).round(4))
print("Total pilot cost (USD):", round(log["cost_usd"].sum(), 2))
print("Hours saved:", round(log["minutes_saved"].sum() / 60))
print(log["agent_outcome"].value_counts(normalize=True).round(3))
```

Column	Meaning
event_id, timestamp, date	One AI run and when it happened
agent_id, team	Who used it (120 agents in 4 teams); drives adoption measures
use_case, model	Triage & routing (Haiku 5.5), Draft reply and Policy knowledge Q&A (Sonnet 5.5)
input_tokens, cached_tokens, output_tokens, cost_usd	Usage and cost of the run (simplified caching: no cache-write premium)

Column	Meaning
latency_ms	Time to answer, in milliseconds
agent_outcome	Accepted, Edited or Rejected by the agent — your quality signal
minutes_saved	Full saving if accepted, half if edited, −0.5 min if rejected (time wasted reading a bad answer)
guardrail_flag	No flag, PII blocked, Low confidence, Off-policy
escalated_to_human	Yes / No: the case was sent to a supervisor

STEP 2 Load and type the data in Power Query

1. Power BI Desktop → Get data → Text/CSV → `ai_event_log.csv` → **Transform data**.
2. Rename the query **FactAIEvents**. Set types: date → Date; timestamp → Date/Time; cost_usd and minutes_saved → Decimal number; token columns and latency_ms → Whole number; everything else → Text.
3. Load `baseline_targets.csv` as a second query named **Targets**.
4. Right-click FactAIEvents → **Reference**. Keep only the team column, Remove duplicates, rename the query **DimTeam**. Do the same for use_case → **DimUseCase**.
5. Close & Apply.

STEP 3 Build the star schema

A star schema has one fact table (events) in the middle and small dimension tables around it that you filter by. Create the date table with DAX (Modeling → New table), then mark it as a date table (Table tools → Mark as date table → Date).

DimDate (calculated table)

```
DimDate =
ADDCOLUMNS (
    CALENDAR ( DATE ( 2026, 1, 1 ), DATE ( 2026, 4, 30 ) ),
    "Month", FORMAT ( [Date], "YYYY-MM" ),
    "Week Start", [Date] - WEEKDAY ( [Date], 2 ) + 1      -- Monday of each week
)
```

From (one side)	To (many side)	Cardinality / direction
DimDate[Date]	FactAIEvents[date]	One-to-many, single
DimTeam[team]	FactAIEvents[team]	One-to-many, single
DimUseCase[use_case]	FactAIEvents[use_case]	One-to-many, single
Targets	(no relationship)	Used only in cards and reference lines

Then hide the date, team and use_case columns in FactAIEvents so report builders always slice by the dimension tables.

STEP 4 Write the DAX measures

Create an empty table called **_Measures** (Enter data → OK) and put every measure in it. Each line below is one measure: select _Measures → New measure → paste one line (the text before "=" is the name). Lines starting with -- are just headings. Assumptions come from A2 so the dashboard and the business case use the same numbers.

_Measures

```
-- Assumptions (same as A2) ------------------------------------------
FX INR per USD      = 88
Agent Cost per Hour = 400
Realisation         = 0.5
Pilot Agents        = 120

-- Cost and value ----------------------------------------------------
AI Runs             = COUNTROWS ( FactAIEvents )
AI Cost INR         = SUM ( FactAIEvents[cost_usd] ) * [FX INR per USD]
Cost per Run INR    = DIVIDE ( [AI Cost INR], [AI Runs] )
Hours Saved         = SUM ( FactAIEvents[minutes_saved] ) / 60
Realised Value INR  = [Hours Saved] * [Agent Cost per Hour] * [Realisation]
Net Value INR       = [Realised Value INR] - [AI Cost INR]
Cumulative Net Value INR =
    VAR LastDay = MAX ( DimDate[Date] )
    RETURN CALCULATE ( [Net Value INR], FILTER ( ALL ( DimDate ), DimDate[Date] <= LastDay ) )

-- Quality and risk --------------------------------------------------
Acceptance Rate     = DIVIDE ( CALCULATE ( [AI Runs],
                          FactAIEvents[agent_outcome] IN { "Accepted", "Edited" } ), [AI Runs] )
Override Rate       = DIVIDE ( CALCULATE ( [AI Runs], FactAIEvents[agent_outcome] = "Rejected" ), [AI Runs] )
Guardrail Flag Rate = DIVIDE ( CALCULATE ( [AI Runs], FactAIEvents[guardrail_flag] <> "No flag" ), [AI Runs] )
PII Blocks          = CALCULATE ( [AI Runs], FactAIEvents[guardrail_flag] = "PII blocked" )
Human Escalation Rate = DIVIDE ( CALCULATE ( [AI Runs],
```

_Measures (continued)

```
                          FactAIEvents[escalated_to_human] = "Yes" ), [AI Runs] )
P95 Latency (s)     = PERCENTILE.INC ( FactAIEvents[latency_ms], 0.95 ) / 1000

-- Adoption ----------------------------------------------------------
Active Agents       = DISTINCTCOUNT ( FactAIEvents[agent_id] )
Adoption %          = DIVIDE ( [Active Agents], [Pilot Agents] )
Runs per Agent      = DIVIDE ( [AI Runs], [Active Agents] )
Runs per Day        = DIVIDE ( [AI Runs], DISTINCTCOUNT ( FactAIEvents[date] ) )
Teams Live          = DISTINCTCOUNT ( FactAIEvents[team] )
```

- **Why FILTER(ALL(DimDate), ...)** in the cumulative measure: it clears every date filter (day, week or month on the axis) and then keeps all days up to the last visible one. Using only `DimDate[Date] <= LastDay` breaks when the axis is Week Start.
- **Why DIVIDE** instead of "/": it returns blank instead of an error when there are no runs (for example, a team with no data on a day).
- **Adoption %** divides by all 120 agents, so read it at total level; when sliced by one team, compare against that team's 30 agents.
- **Cost per resolution** needs the number of complaints closed, which comes from the CRM, not the AI log: `DIVIDE ( [AI Cost INR], [Complaints Closed] )`. In the simulation, Cost per Run is the stand-in; say so in the KPI dictionary.

Put the measures in a card visual with no filters and check them against these values for the full 90 days:

Measure	Expected	Measure	Expected
AI Runs	64,588	Acceptance Rate	89.7%
AI Cost INR	≈ ₹45,017	Override Rate	10.3%
Cost per Run INR	≈ ₹0.70	Guardrail Flag Rate	3.4%
Hours Saved	≈ 3,370	PII Blocks	944
Realised Value INR	≈ ₹6.74 lakh	Human Escalation Rate	6.0%
Net Value INR	≈ ₹6.29 lakh	P95 Latency (s)	≈ 2.89
Active Agents / Teams Live	120 / 4	Runs per Day / per Agent	≈ 718 / 538

Say this before anyone else does

Net value here covers **model usage only**. It leaves out the run team, hosting and the build cost, which are in the A2 business case. Add a text box on the CFO page that says so. A dashboard that silently flatters the AI is exactly what a CFO distrusts.

STEP 5 Build the four pages

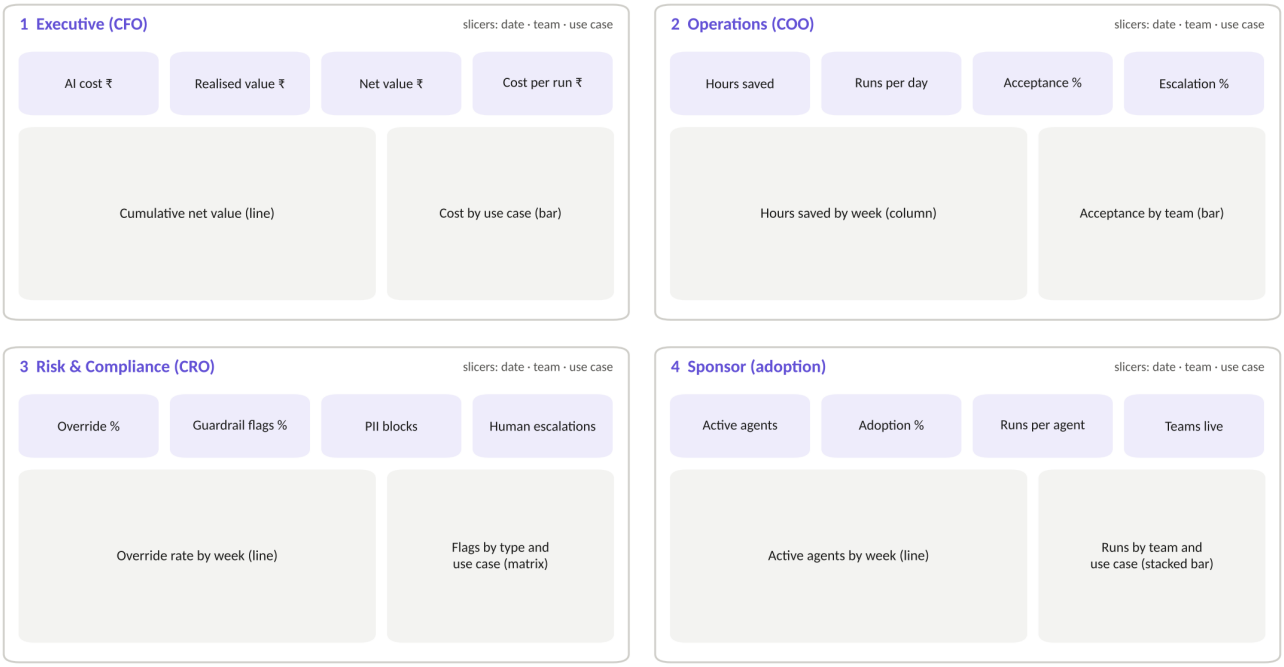


Figure 5. Page layouts. Four cards across the top answer "how are we doing?"; the big chart answers "which way is it going?"; the side chart answers "where?".

Page	Audience and their question	Visuals
1 Executive	CFO: "Is this paying for itself?"	Cards: AI Cost INR, Realised Value INR, Net Value INR, Cost per Run INR · line: Cumulative Net Value INR by Date · bar: AI Cost INR by use case · note on what costs are excluded
2 Operations	COO: "Is it making the team faster?"	Cards: Hours Saved, Runs per Day, Acceptance Rate, Human Escalation Rate · column: Hours Saved by Week Start · bar: Acceptance Rate by team · Targets card: AHT 25 → 19 min
3 Risk & Compliance	CRO / Compliance: "Is it safe?"	Cards: Override Rate, Guardrail Flag Rate, PII Blocks, Human Escalation Rate · line: Override Rate by Week Start with a 15% threshold line · matrix: guardrail_flag × use case
4 Sponsor	Head of Customer Service: "Are people using it?"	Cards: Active Agents, Adoption %, Runs per Agent, Teams Live · line: Active Agents by Week Start (rises from 52 to 120) · stacked bar: AI Runs by team and use case

STEP 6 Apply the design rules and add the notes

- **Page titles as questions** ("Is it safe?"), so each page has one job.
- **At most four cards** per page. Format rates as %, money as ₹ with thousands separators.
- **Same colours everywhere**: one colour per use case on every page.
- **Sync slicers** (View → Sync slicers) for date, team and use case across all four pages.
- **"How to read this page"** text box, two lines, bottom left. Example: "Override rate is the share of AI answers the agent rejected. Above 15% for two weeks triggers a review (see governance plan)."
- **KPI dictionary** as a fifth, hidden-in-presentation page or a separate PDF: name, definition, DAX, owner, threshold.

> **"How do you calculate deflection rate?" — the interview answer**
>
> Deflection rate = contacts fully resolved by self-service (for example a customer-facing assistant) ÷ all contacts that started in that channel. Kestrel Bank's Phase 1–2 tool is an **agent assistant**, so nothing is deflected — every complaint still reaches an agent. If a customer-facing assistant is added later, add a measure such as `DIVIDE ( [Self-served Resolved], [Self-service Sessions] )` and watch the reopen rate alongside it, because false deflection (customer gives up) looks like success.

STEP 7 Save and share it

1. Save `A3_AI_Value.pbix`. File → Export → **Export to PDF** for a version anyone can open.
2. Take one clean screenshot per page for your case study and LinkedIn post.
3. Publishing to the Power BI service needs a work or school account, and Publish to web needs a Pro licence [13]. For a portfolio, sharing the .pbix and the PDF is enough — and it avoids putting even simulated bank data on a public link.

Done when...

- ☐ Simulated data generated; the cover page says it is simulated
- ☐ Star schema with DimDate marked as a date table and three one-to-many relationships
- ☐ All measures created; values match the Step 4 check table
- ☐ Four pages, each answering one audience's question, with notes and synced slicers
- ☐ KPI dictionary written
- ☐ .pbix and PDF export saved in `03_A3_dashboard`

FLAGSHIP PROJECT · WEEKS 7–12

A Flagship A — AI Transformation Blueprint

Flagship A at a glance	
Objective	Act as the consultant hired to answer: "Where should Kestrel Bank use AI in complaint handling, what will it return, and how do we roll it out safely?"
Builds on	A1, A2, A3 · #4 CRM BA Capstone (BRD, user stories, RTM) · #3 Client Feedback Automation (Phase 0 quick win) · #5 Project NOVA (roadmap in MS Project) · #2 Customer Segmentation (optional theme clustering).
Deliverables	(1) Discovery summary: stakeholder map, interview guide, pain-point register with baselines. (2) As-is and to-be BPMN with every step tagged H / R / AI / AG. (3) Use-case shortlist on a value-vs-feasibility matrix, including what not to automate. (4) Business case with hard-only and full views. (5) Governance plan: DPDP and FREE-AI control map, risk register, human-review points. (6) Phased roadmap and a 10-slide executive deck with go / no-go criteria. Plus a 2-page case study.
Skills shown	Engagement framing, discovery, process redesign, requirements, prioritisation, business case, AI governance, roadmapping, executive storytelling.

STEP 1 Frame the engagement (week 7)

Consultants start with the question, not the answer. Write the **SCQ** (Situation, Complication, Question) and an **issue tree** that breaks the question into parts you can answer with evidence. Every later step answers one branch.

Engagement framing (save as 04_FlagshipA/00_framing.md)

```
Situation:    Kestrel Bank handles ~40,000 complaints a month across 4 teams.
Complication: Volumes are rising, agents spend long periods searching for policy, and
              customers can go to the RBI Ombudsman if there is no reply within 30 days.
Question:     Where should the bank use AI in complaint handling, what will it return,
              and how should it be rolled out safely?

Issue tree
1. Where is value lost today?                  -> A1 diagnostic, interviews (Steps 2-3)
   1.1 Which issues and steps drive volume and delay?
   1.2 Which causes need a process fix, not technology?
2. Which AI use cases are worth doing?         -> Step 5
   2.1 What is the simplest tool that works for each pain?
   2.2 How do they score on value and feasibility?
3. What is it worth?                           -> A2, Step 6
   3.1 What does it cost (build, run, tokens)?
   3.2 What do we get (hard and soft), and how sure are we?
4. How do we do it safely?                     -> Steps 4, 7, 8
   4.1 What do DPDP, RBI FREE-AI and RB-IOS require of this design?
   4.2 Where must a human decide?  4.3 How do we phase and gate it?
```

STEP 2 Run discovery: stakeholders, interviews, pain points (week 7)

2a. Stakeholder map

Stakeholder	Role in the decision	Influence / interest	What they care about	How to engage
Head of Customer Service	Sponsor; owns budget	High / High	Escalations, customer satisfaction, cost	Steering meetings; business case
Principal Nodal Officer	Accountable for grievance redress to RBI	High / High	30-day timeline, Ombudsman cases	Interview; governance plan
Chief Risk / Compliance Officer	Can stop the project	High / Medium	DPDP, FREE-AI, model risk	Early review of control map
CIO / IT architecture	Builds and runs it	High / Medium	Integration, security, cost	Requirements and NFR workshop
Complaint team leads	Change their team's work	Medium / High	Workload, quality checks	Interviews; pilot design
Complaint agents	Daily users	Low / High	Less searching, fewer rewrites, not being replaced	Interviews; champions

Stakeholder	Role in the decision	Influence / interest	What they care about	How to engage
CFO / Finance	Approves investment	High / Low	Payback, realisation of savings	Business-case review

2b. Interview guide (30 minutes)

Try to speak to **three to five real people** who work in bank operations, customer service or compliance — alumni, LinkedIn, family network. Ask for 20–30 minutes, explain it is a learning project, and never ask for confidential data. Even three conversations let you replace "assumed" with "validated in interviews" on your key numbers.

Section	Questions
Warm-up (3 min)	What is your role? How does a complaint reach you?
Process (10 min)	Walk me through the last complaint you handled, step by step. Where did you wait? Where did you search for something? How many people touched it?
Numbers (7 min)	Roughly how long does a typical complaint take you to handle? What share go past 30 days? What share are reopened? Which three complaint types take the most time?
Pain and ideas (7 min)	If you could fix one thing, what would it be? Where would a tool that finds the right policy help? Where would you *not* trust an AI suggestion?
Close (3 min)	What did I not ask that I should have? Who else should I talk to?

2c. Pain-point register

ID	Pain point	Evidence	Who feels it	Frequency	Impact	Baseline metric
P1	Searching four systems for policy and RBI rules	A1 tags: [x]% Knowledge; 4 of 5 interviews	Agents	Every complaint	High	AHT 25 min (assumed)
P2	Complaints routed to the wrong team first	A1 tags: [x]% Triage; interviews	Agents, customers	~1 in [5]	Medium	Re-route rate (assumed)
P3	Writing long replies from scratch	Interviews	Agents	Every complaint	Medium	Reply drafting time
P4	Complaints drifting past 30 days	A1 control gap; interviews	Nodal officer	[12]% (assumed)	High	% resolved ≤ 30 days
P5	Repeat complaints from one product issue	A1 Pareto + relief rate	Customers	Top issue type	High	Relief rate by issue

STEP 3 Redesign the process: as-is to to-be BPMN (week 8)

Start from the A1 as-is map. Apply the non-AI fixes first (acknowledgement rule, categories, day-25 alert), then add AI only where a pain point needs it. Tag every step and show where a human decides.

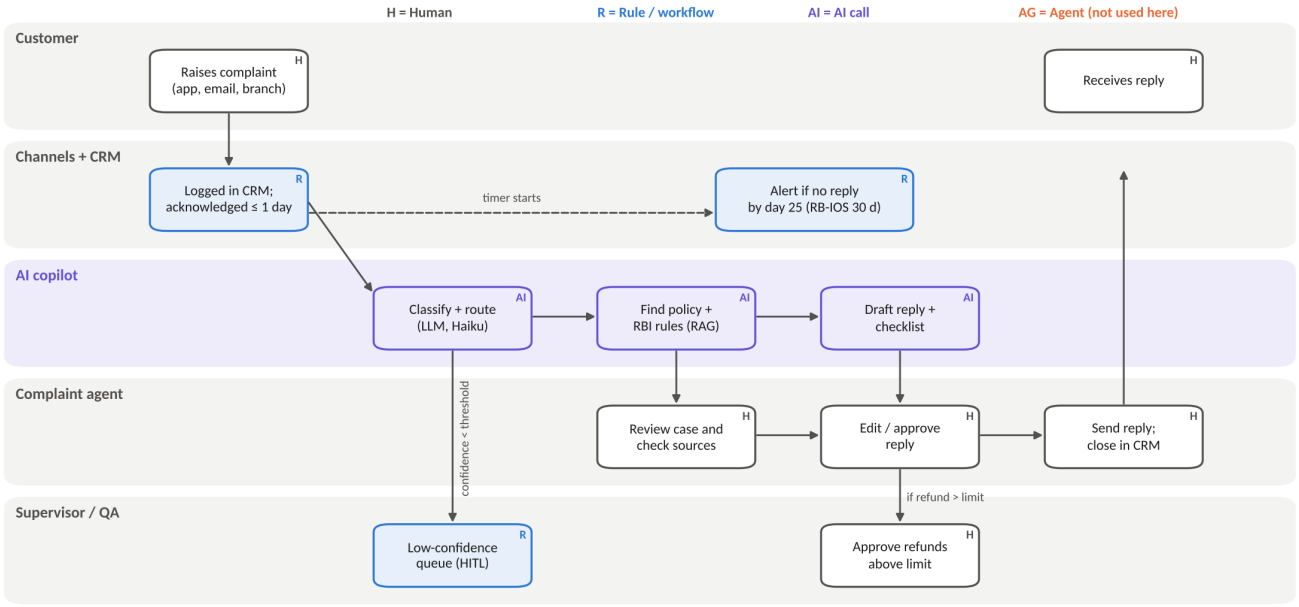


Figure 6. To-be process for Kestrel Bank. AI assists the agent; humans approve every reply, review low-confidence cases and approve refunds above the limit. No step is an autonomous agent in this phase.

- **Count the steps** by tag in both maps (for example as-is: 11 H, 1 R; to-be: 7 H, 4 R, 3 AI) and put the count on the slide. It shows change at a glance.
- **Mark every human-in-the-loop (HITL) point** and write its rule: “Agent approves every outgoing reply”, “Confidence below threshold → supervisor queue”, “Refund above limit → supervisor approval”.
- **Mark the AG tag as unused on purpose.** Saying “no autonomous agent yet, and here is why” is a strong signal of judgement.

STEP 4 Write AI-aware requirements: BRD, user stories, RTM (week 8)

Reuse the templates from your CRM BA Capstone (#4). What changes for AI is that requirements must say how good is good enough, what happens when the AI is wrong, and how it will be checked.

BRD section	AI-specific content to add
1 Objectives and scope	Link each objective to a pain point (P1–P5) and a KPI from A1 / A3.
2 Current and future state	As-is and to-be maps from Step 3.
3 Functional requirements	Per use case: inputs, outputs, users, systems touched.
4 AI quality requirements	Accuracy or acceptance thresholds, citation required for policy answers, refusal when no source is found.
5 Human oversight	HITL rules from Step 3; who can override; how overrides are logged.
6 Data requirements	Data sources, PII masking before any model call, retention, India data residency preference.
7 Non-functional	Latency (P95 under 5 s for knowledge answers), availability, audit logging, cost per run ceiling.
8 Monitoring and acceptance	A3 measures and thresholds; evaluation test set; sign-off owners.

User stories with evaluation criteria (examples)

ID	User story	Acceptance / evaluation criteria
US-01	As a complaint agent, I want the complaint category and team suggested, so that I stop re-routing complaints.	Suggested team correct on ≥ 90% of a 200-complaint labelled test set; agent can change it in one click; changes are logged.
US-02	As a complaint agent, I want to ask a policy question and get an answer with the source, so that I stop searching four systems.	Answer cites the policy or circular section; says “not found” rather than guessing when no source exists; P95 latency under 5 s; ≥ 85% of answers rated correct by two reviewers on a 100-question test set.

ID	User story	Acceptance / evaluation criteria
US-03	As a supervisor, I want every AI-drafted reply to need agent approval before sending, so that no unreviewed text reaches a customer.	No send path without approval (tested); every reply stores who approved it and whether it was edited.

Requirements traceability matrix (RTM)

Req ID	Pain point	User story	Design element	Test / evaluation	KPI (A3)	Status
FR-01	P2	US-01	Classify + route (Haiku)	Labelled test set, 200 items	Acceptance Rate	Draft
FR-02	P1	US-02	Policy Q&A (RAG) with citations	100-question test set	Override Rate	Draft
FR-03	P3	US-03	Approval step before send	Negative test: try to send unapproved	Human Escalation Rate	Draft
NFR-01	—	—	PII masking before model call	Test with 50 synthetic PII examples	PII Blocks	Draft

STEP 5 Prioritise the use cases (week 9)

Build a long list from A1 (Step 11), the pain-point register and the interviews. Then, for each one, climb the **decision ladder** and stop at the first rung that solves the problem: **process fix → rules → classical ML → LLM / RAG → agent**. The cheapest rung that works is the right answer.

#	Use case	Lowest rung that works	Why
1	Triage and routing	Small LLM now; classical ML later	CRM has no consistent categories yet (A1), so there are no labels to train on. Once 6 months of labelled data exist, test a cheaper classifier.
2	Policy knowledge Q&A	LLM + RAG	Answers must come from changing documents, with citations.
3	Draft reply, human approves	LLM + human approval	Free-text writing; risk controlled by approval.
4	Escalation summary	LLM	Summarising long case histories.
5	Vulnerable-customer flag	Rules + classical ML + human	Sensitive; needs clear criteria and fairness checks first.
6	Root-cause theme mining	Classical ML (batch)	TF-IDF + K-Means as in A1; an LLM can name the themes. Monthly batch run.
7	Auto-send replies with no review	Not recommended	Customer-facing text with regulatory weight; failure cost too high.
8	AI decides compensation	Rules + human	Approval limits are a policy question; keep the decision human.
9	Feedback sentiment flow (#3)	Prebuilt AI (Power Automate + AI Builder)	Already built in your portfolio; quick win.

Next, score each use case 1–5 on six criteria. A use case **passes** if it scores at least 20 out of 30 and no criterion scores 1. The six criteria are financial value, customer / regulatory impact, data readiness, technical feasibility, risk (5 = lowest risk) and adoption readiness. Value = average of the first two; feasibility = average of the last four.

#	Use case	Fin. value	Cust. / reg.	Data	Tech.	Risk	Adopt.	Total	Result
1	Triage and routing	4	4	5	5	4	4	26	Pass
2	Policy knowledge Q&A	4	4	4	4	4	4	24	Pass
3	Draft reply, human approves	5	4	4	3	3	3	22	Pass
4	Escalation summary	3	3	5	5	4	4	24	Pass
5	Vulnerable-customer flag	3	5	3	3	2	3	19	Not yet
6	Root-cause theme mining	3	3	4	4	5	4	23	Pass
7	Auto-send replies	5	2	4	3	1	2	17	Reject
8	AI decides compensation	4	2	3	2	1	2	14	Reject
9	Feedback sentiment flow	2	3	4	5	5	4	23	Pass

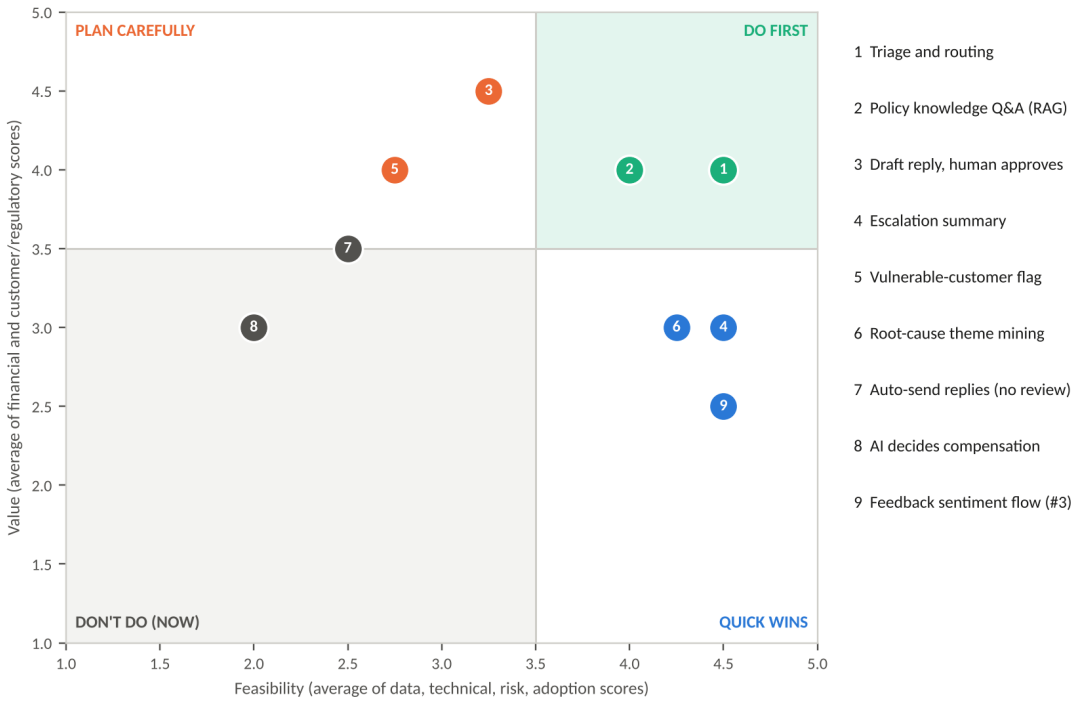


Figure 7. Value-vs-feasibility matrix from the scores above. Do first: 1 and 2. Plan carefully: 3 (Phase 2, with approval) and 5. Quick wins: 4, 6, 9. Don't do now: 7 and 8.

"Why didn't you recommend an agent?" — your answer

"Because the decision ladder stopped earlier. Every high-value step is either a lookup (RAG), a draft a human approves, or a decision the bank's policy says a person must make. An agent adds autonomy, and autonomy adds risk, without solving a pain point we found. Phase 3 revisits it for narrow internal steps, such as gathering case documents, which is where Flagship B starts."

STEP 6 Present the business case (week 9)

Take the A2 Base, Worst and Best results onto one slide. Lead with the decision, not the spreadsheet.

	Hard benefits only	Full view (incl. escalations avoided)
NPV at 12%, 3 years (Base)	−₹23.4 lakh	₹127.5 lakh
Payback (Base)	Not within 3 years	18 months
Range across scenarios (NPV)	−₹123.3 lakh to ₹75.6 lakh	−₹68.0 lakh to ₹324.6 lakh
Biggest drivers	Adoption, realisation factor	Adoption, cost per escalation

Suggested slide headline

"A ₹86 lakh build pays back in about 18 months — if the pilot proves fewer Ombudsman escalations and at least 60% adoption. We recommend funding Phase 1 only and deciding on Phase 2 at Gate 2."

Note: this is a stage-gated recommendation, not a blanket "yes". That is how Big 4 business cases are usually framed when the benefit depends on behaviour change.

STEP 7 Write the governance plan (week 10)

Three Indian frameworks shape this design. Keep the summary factual and cite it; you are mapping requirements to controls, not giving legal advice.

- **DPDP Act and Rules:** the Rules were notified on 13 November 2025 and take effect in phases, with most obligations from May 2027 [9]. Breaches must be reported to the Data Protection Board without delay, with a detailed report within 72 hours [10]. Design for these now; the system will be live before May 2027.
- **RBI FREE-AI framework (August 2025):** the committee set out 7 principles ("Sutras") and 26 recommendations, including a board-approved AI policy, an inventory of AI systems, disclosure to customers when they deal with AI, and strong grievance redress [7,8].
- **RB-IOS 2026:** customers can go to the RBI Ombudsman when there is no reply within 30 days, and the Ombudsman can award up to ₹30 lakh for consequential loss [6]. This is why reply quality and the 30-day clock are governance issues, not just service ones.

7a. Control map

Requirement (source)	What it means for this design	Control	Owner	Evidence
DPDP: purpose limitation and data minimisation	Send the model only what the task needs	Mask PII (names, account and card numbers, PAN) before every model call; log masked fields	IT security	PII Blocks measure; masking test set
DPDP: security safeguards; breach reporting	Logs and prompts are personal data	Encryption, access control, retention limit on logs; breach runbook with 72-hour report step	CISO / DPO	Runbook; access review
DPDP: processor obligations	The model provider processes data for the bank	Contract terms on use of data, deletion, sub-processors; India region where available	Procurement + Legal	Signed contract
FREE-AI: AI policy and inventory	The copilot must be on the bank's AI register	Add each use case to the AI inventory with owner, risk tier, model, test results	Chief Risk Officer	Inventory entry
FREE-AI: disclosure and grievance redress	Customers must not be misled about AI use	Human approves every reply; disclosure wording agreed with Compliance for any customer-facing AI	Compliance	Approval logs
RB-IOS 2026: 30-day timeline	Faster replies must not lower quality	Day-25 alert; QA sample of AI-assisted replies each week	Nodal officer	A1 control plan KPIs
Model risk (good practice)	Quality can drift as policies change	Re-run evaluation set monthly and after each policy change; override threshold 15%	AI product owner	A3 Override Rate trend

7b. Risk register

Risk	Likelihood	Impact	Mitigation	Early warning (A3)
Wrong policy answer leads to wrong reply	Medium	High	Citations required; agent approval; monthly evaluation	Override Rate > 15%
Personal data sent to the model	Low	High	Masking before calls; red-team test set	PII Blocks spike
Agents do not adopt the tool	Medium	High	Champions, training, workflow fit, listen to feedback	Active Agents flat
Benefits not realised as money	Medium	Medium	Agree with Finance how saved hours are used (hiring freeze, overtime)	Hours Saved vs plan
Cost overrun on tokens	Low	Low	Routing, caching, monthly cost ceiling alert	Cost per Run INR
Vendor or model change breaks quality	Medium	Medium	Pin model versions; re-test before upgrade	Acceptance Rate drop

7c. RACI

Activity	Sponsor (Head CS)	AI product owner	IT / engineering	Compliance / Risk	Team leads
Approve use cases and budget	A	R	C	C	I
Build and test	I	A	R	C	C
Approve go-live at each gate	A	R	C	R	C
Monitor KPIs (A3)	I	A	R	C	R
Handle incidents and breaches	I	R	R	A	I

R = responsible (does it), A = accountable (one per row, signs off), C = consulted, I = informed.

STEP 8 Build the roadmap with stage gates (week 11)

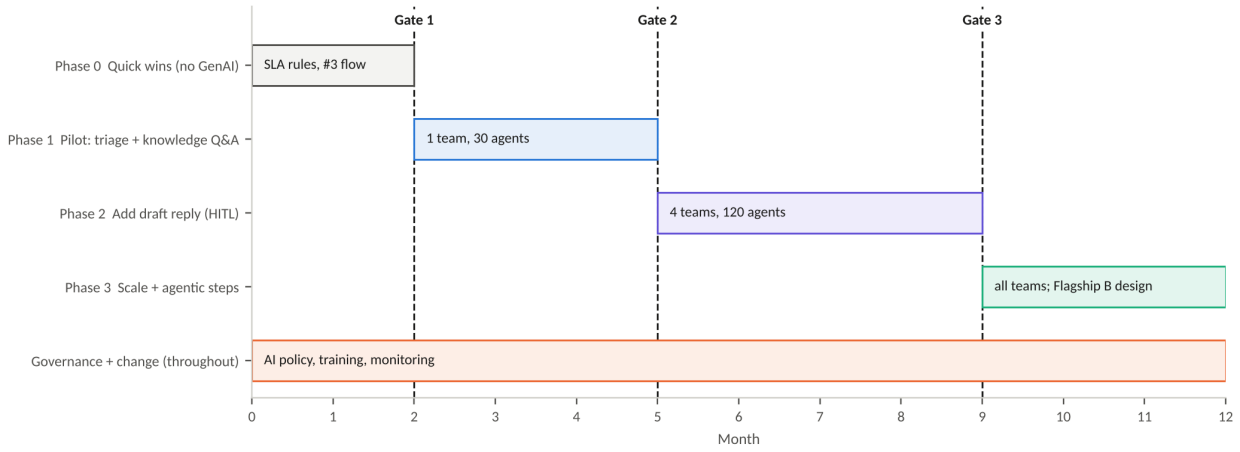


Figure 8. Twelve-month roadmap. Phase 0 uses no GenAI at all; each gate is a go / no-go decision with criteria agreed in advance.

Gate	Decision	Go criteria (all must be met)
Gate 1 (month 2)	Start the pilot	Phase 0 fixes live · privacy review done · AI inventory entry approved · evaluation sets ready (200 triage, 100 policy questions) · baseline KPIs captured for the pilot team
Gate 2 (month 5)	Add draft reply; expand to 4 teams	Acceptance ≥ 80% · Override ≤ 15% · zero unmasked PII incidents · ≥ 60% of pilot agents active weekly · escalations trending down vs baseline · business case re-run with pilot data
Gate 3 (month 9)	Scale to all teams; design agentic steps	Gate 2 criteria held for 3 months · payback ≤ 24 months on measured data · Compliance sign-off on any customer-facing change

Building it in MS Project or ProjectLibre (reuse your Project NOVA (#5) skills):

1. Enter phases as summary tasks and the work under each as sub-tasks with durations in weeks.
2. Add gates as milestones (duration 0) and link tasks finish-to-start so a gate cannot pass before its inputs are done.
3. Assign roles (not names) as resources: AI product owner, engineer, compliance reviewer, trainer.
4. Show the Gantt view, add the critical path, export as PNG/PDF for the deck. ProjectLibre is free if you have no MS Project licence.

> **Phase 0 quick win: reuse your portfolio project #3**
>
> Your Client Feedback Automation (Power Automate + AI Builder) becomes Phase 0: collect post-resolution feedback, score sentiment, and route negative feedback to the team lead within a day. It uses prebuilt AI, needs no GenAI approval, and gives the sponsor a visible early result.

Change management belongs on the roadmap, not in an appendix: one champion per team, 2-hour training before go-live, a weekly 15-minute feedback huddle during the pilot, and a clear message that the tool removes searching and typing, not jobs.

STEP 9 Build the 10-slide executive deck (week 11)

Every slide title is an **action title**: a full sentence stating the takeaway. Someone reading only the titles should get the whole story.

#	Action title (example)	Content	From
1	AI can cut complaint handling effort by about a quarter, if we start with the process	Executive summary: 3 findings, 1 recommendation, the ask	All
2	A handful of issue types and one root cause drive most complaints	Pareto, relief rate, top root causes	A1
3	Six fixes need no AI and should start now	Non-AI fix list with owners	A1
4	Agents lose most time searching for policy and rewriting replies	Pain-point register, interview quotes	Step 2
5	AI should assist agents at three points, with a human approving every reply	To-be BPMN with tags and HITL points	Step 3
6	Two use cases are ready now; two should not be done at all	Matrix + scorecard summary	Step 5
7	The base case pays back in 18 months, but only if escalations fall	Hard vs full table, tornado	A2, Step 6
8	The design meets DPDP, FREE-AI and RB-IOS requirements from day one	Control map summary, top risks	Step 7
9	Three gates let the bank stop or scale based on evidence	Roadmap, gate criteria	Step 8
10	We will know it is working within 90 days	Dashboard screenshots, KPI thresholds, next steps and the ask	A3

- One message per slide; at most about 40 words of body text; the chart does the talking.
- Put sources and “measured / assumed” labels in small text at the bottom of each slide.
- Rehearse the deck in 8 minutes. Interviewers often say “walk me through it in five”.

STEP 10 Write the case study and package everything (week 12)

Two-page case study structure

Section	What to write (about)
Context (60 words)	Fictional mid-size Indian bank; public CFPB data as a stand-in; your role as consultant.
Problem (60 words)	The SCQ from Step 1.
Approach (150 words)	Diagnostic → discovery → redesign → prioritisation → business case → governance → roadmap. Name the tools.
Findings (150 words)	Three findings with numbers, each labelled measured or assumed.
Recommendation (100 words)	What to do, what not to do, and the gated plan.
Impact and limits (80 words)	Expected outcomes; what the data could not tell you; what you would validate next.
Visuals	Pareto, to-be map, matrix, tornado, one dashboard page.

Packaging checklist

- **README.md** at the top of the repository or Drive folder: one paragraph on the engagement, five key findings, a map of the folders, and “how to reproduce” (Python versions, run order).
- **PDFs of everything** (deck, one-pagers, dashboard, case study): reviewers will not install software to look at your work.
- **LinkedIn post**: one image (the to-be map or matrix), three findings, one lesson (“the best AI recommendation was the six fixes that needed no AI”), link.
- **Interview story** in STAR form, 90 seconds: Situation (bank, complaints), Task (where should AI go), Action (diagnostic, scoring, business case, governance), Result (gated plan, 18-month payback conditional on escalations, two use cases rejected and why).

Done when...

- ☐ Framing note with SCQ and issue tree
- ☐ Stakeholder map, interview guide, notes from at least three conversations, pain-point register
- ☐ As-is and to-be BPMN with tags and HITL rules
- ☐ BRD with AI sections, user stories with evaluation criteria, RTM
- ☐ Decision ladder, scorecard and matrix, including what not to automate
- ☐ Business-case slide with hard-only and full views
- ☐ Governance plan: control map, risk register, RACI
- ☐ Roadmap with gates and criteria (MS Project / ProjectLibre export)
- ☐ 10-slide deck with action titles; 2-page case study; README; LinkedIn post

QUALITY REVIEW

Final Check: Does It Meet the Objectives?

Use this section twice: once when you plan, and again before you show anyone the work. The first table maps every deliverable promised in the Flagship Projects document to the step in this guide that builds it and the file it ends up in.

Deliverables traceability

Project	Promised deliverable	Built in	File
A1	SIPOC	A1 Step 2	A1_onepager.pdf
A1	As-is process map with “where AI could help” tags	A1 Steps 9, 11	as_is.drawio / .png
A1	Pareto of issue types	A1 Step 6	pareto_issues.png
A1	Fishbone (and 5 Whys)	A1 Step 8	fishbone.png
A1	Baseline KPI table	A1 Step 5	baseline_kpis.csv
A1	List of non-AI fixes; control KPIs	A1 Steps 10, 12	A1_onepager.pdf
A2	Input sheet; cost-per-query calculator	A2 Steps 2–3	A2_GenAI_Cost_ROI.xlsx
A2	Three-year cash flow with ramp-up and NPV	A2 Steps 4–5	same
A2	Best / base / worst scenarios	A2 Steps 2, 6	same
A2	Tornado chart; one-page summary	A2 Steps 7–8	same + A2_one_pager.pdf
A3	Star-schema model	A3 Steps 2–3	A3_AI_Value.pbix
A3	6–8 DAX measures (guide gives 18, plus 4 assumption measures)	A3 Step 4	same + KPI dictionary
A3	Pages for CFO, COO, Risk and sponsor	A3 Steps 5–6	A3_pages.pdf
FA	(1) Discovery summary: stakeholder map, interview guide, pain-point register	Flagship Steps 1–2	discovery/
FA	(2) As-is and to-be BPMN, every step tagged H / R / AI / AG	Flagship Step 3 (+ A1 Step 9)	process/
FA	(3) Use-case shortlist on value-vs-feasibility matrix, incl. what not to automate	Flagship Step 5	process/ + deck
FA	(4) Business case, hard-only and full views	Flagship Step 6 (from A2)	deck slide 7
FA	(5) Governance plan: DPDP / FREE-AI control map, risk register, human-review points	Flagship Steps 3, 7	governance/
FA	(6) Phased roadmap and 10-slide deck with go / no-go criteria	Flagship Steps 8–9	roadmap/, deck/
FA	2-page case study	Flagship Step 10	case_study.pdf
Portfolio	#2, #3, #4, #5, #6 reused	A1 Steps 7 (#2), all (#6); Flagship Steps 4 (#4), 8 (#3, #5)	—

Interview questions and where your answer is

Question (from the Flagship Projects document)	Your answer, in short	Where
What was your biggest finding?	Root cause is knowledge and process, not people; most fixes need no AI.	A1 Steps 8, 13
Which fixes needed no technology?	Acknowledgement SLA, categories, approval limits, knowledge owner, day-25 alert, product fix.	A1 Step 10
How did you measure the baseline?	CFPB data for mix and rates; volumes and times as labelled assumptions; no resolution date in the data.	A1 Step 5, 0.5
Estimate monthly LLM cost for 2,000 users.	2,000 × 10 × 22 × $0.017 ≈ $7,500 (≈ ₹6.6 lakh).	A2 Step 8
What is your riskiest assumption?	Adoption — the widest bar on the tornado.	A2 Step 7

Question (from the Flagship Projects document)	Your answer, in short	Where
What happens to your ROI if adoption halves?	From 80% to 40%: full-view NPV falls from ₹127.5 lakh to about −₹27 lakh; no payback in 3 years.	A2 model
How would you cut the cost?	Routing, caching, batch for non-urgent work, output caps — but adoption is the biggest lever.	A2 Step 8
How do you calculate deflection rate?	Self-served resolved ÷ self-service contacts; not applicable to an agent assistant.	A3 Step 6
How would you show ROI to a CFO?	Cumulative net value, cost per run, and a note on excluded costs; full case in A2.	A3 Step 5
Which measure would warn you first?	Override Rate by week — quality problems show there before cost or hours move.	A3 Step 5
How did you prioritise use cases?	Decision ladder, six-criterion scorecard with pass rule, matrix.	Flagship Step 5
Why didn't you recommend an agent?	The ladder stopped earlier; autonomy adds risk without solving a found pain point.	Flagship Step 5
How does DPDP change your design?	PII masked before model calls, processor contract, log retention, breach runbook.	Flagship Step 7

Quality checklist before you show it

- Every number is labelled **measured**, **sourced** or **assumed**, and every assumption is in the register.
- The client is clearly fictional and the data clearly a stand-in; the A3 data is clearly simulated.
- Every slide and one-pager has an action title that states the finding.
- The same assumptions appear in A2, A3 and the deck (FX ₹88, ₹400 per hour, 50% realisation, 120 agents).
- At least one recommendation says **not** to use AI, with a reason.
- Someone outside your field can read the deck titles alone and understand the story.
- Code runs from a clean folder in the order given; the README says how.

Common mistakes that cost credibility

Mistake	Do this instead
Reporting resolution time from CFPB data	State that the data has no resolution date; use interviews and label assumptions.
One ROI number with no range	Show Worst / Base / Best and the tornado; separate hard from soft benefits.
Leading with the model or tool	Lead with the business problem and the non-AI fixes; tools come last.
"AI will replace agents"	AI removes searching and typing; benefits depend on how saved time is used (realisation factor).
Dashboard of activity (runs, tokens)	Dashboard of outcomes (hours saved, acceptance, overrides, adoption) per audience.
Governance as a final slide	Governance shapes the design: masking, approval steps, gates.
Quoting prices without a date	Every price carries a date and a source; prices change often.

CHECKED 7 NOVEMBER 2026

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Notes. Kestrel Bank is fictional. CFPB data is US public complaint data used only as a stand-in for the shape of the problem. Business-case figures come from the A2 model in this guide and depend on the stated assumptions. Prices are list prices on the date above. Regulatory summaries are for project design, not legal advice.